

Dynamic Guideline: WHO-Based COVID-19 - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Latest WHO COVID-19 treatment guideline identified as of 29 April 2026

- **Guideline title:** *Therapeutics and COVID-19: living guideline, August 2025*
- **WHO publication date:** 29 September 2025; MAGICapp version listed as 1 October 2025
- **Status:** WHO 15th version / 14th update; replaces earlier therapeutic versions
- **Target population:** People with COVID-19 across disease severities, with separate recommendations for non-severe, severe, and critical disease
- **Scope and objectives:** Recommendations on therapeutics used to **treat** COVID-19; WHO explicitly separates this from guidance on drugs used to **prevent** COVID-19 and from broader clinical management guidance [WHO Therapeutics August 2025](#), [WHO IRIS PDF](#).

Important currency note: No 2026 WHO COVID-19 guideline update was identified in the available WHO sources. Therefore, as of **29 April 2026**, the latest identified WHO COVID-19 therapeutic guideline remains the **August 2025 living guideline**, published by WHO on **29 September 2025** [WHO Therapeutics August 2025](#).

Exactly 5 WHO guideline or normative documents searched and synthesized

No.	WHO document	Date / status	COVID-19 relevance	Key use in this report
1	<i>Therapeutics and COVID-19: living guideline, August 2025</i>	Published 29 September 2025	Direct COVID-19 treatment guideline	Latest WHO therapeutic recommendations, including metformin update WHO Therapeutics August 2025
2	<i>Clinical management of COVID-19: living guideline, June 2025</i>	Published 15 July 2025	Direct COVID-19 clinical management guideline	Antibiotic stewardship, post-COVID condition, clinical management methodology WHO Clinical Management June 2025
3	WHO living guideline on drugs to prevent COVID-19	Separate WHO living guideline referenced by WHO	COVID-19 prevention pharmacotherapy	Scope distinction; detailed recommendations were not available in the retrieved source excerpts WHO Therapeutics August 2025

No.	WHO document	Date / status	COVID-19 relevance	Key use in this report
4	WHO public health and social measures COVID-19 guideline	WHO IRIS PDF identified	COVID-19 public health measures	Methodology context: GDG, GRADE tables, external review, steering committee WHO PHSM IRIS PDF
5	<i>Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)</i>	Published 11 December 2025	Not COVID-specific; WHO guideline-development methodology document requested in the search scope	Identified as the most recent WHO document matching the requested title pattern; informs inclusivity and stakeholder-engagement principles, not COVID clinical recommendations WHO publication page

Most recent requested WHO guideline-development title identified:

Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022), published **11 December 2025**. WHO states that the brief describes youth engagement through consultations and a youth-led convening in development of the 2022 WHO abortion care guideline; it is not a COVID-19 clinical guideline and should not be used to make COVID-19 treatment decisions [WHO publication page](#), [WHO webinar page](#).

1.2 Key Recommendations

GRADE interpretation used in this report

GRADE domain	Meaning
High quality	Further research is unlikely to change confidence in the effect estimate.
Moderate quality	Further research may change confidence in the effect estimate.
Low quality	Further research is likely to change confidence in the effect estimate.
Very low quality	Effect estimate is very uncertain.
Strong recommendation	Most well-informed patients should receive, or not receive, the intervention.
Conditional recommendation	Different choices may be appropriate depending on patient values, risk, feasibility, access, and clinical context.

WHO-based COVID-19 treatment and management recommendations

Clinical area	Recommendation	Target population	GRADE evidence quality	Strength
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Clinical area	Recommendation	Target population	GRADE evidence quality	Strength
Outpatient antiviral treatment	Use nirmatrelvir/ritonavir for non-severe COVID-19 patients at highest or high risk of hospitalization, when started early and when drug–drug interactions are manageable.	Non-severe COVID-19, high/highest hospitalization risk	Moderate	Strong
Outpatient antiviral treatment	Consider nirmatrelvir/ritonavir for patients at moderate risk of hospitalization after individualized assessment.	Non-severe COVID-19, moderate hospitalization risk	Low to Moderate	Conditional
Outpatient antiviral treatment	Do not routinely use nirmatrelvir/ritonavir in low-risk non-severe COVID-19.	Non-severe COVID-19, low hospitalization risk	Low	Conditional against
Outpatient antiviral treatment	Consider remdesivir as an alternative for high-risk non-severe COVID-19 when feasible.	Non-severe COVID-19, high/highest risk	Moderate	Conditional
Outpatient antiviral treatment	Consider molnupiravir only when preferred options are unavailable, unsuitable, or contraindicated.	Non-severe COVID-19, high-risk patients	Low	Conditional
Outpatient antiviral treatment	Do not use molnupiravir in low-risk non-severe COVID-19.	Non-severe COVID-19, low risk	Low	Strong against
Metformin	Do not use metformin for acute management of COVID-19. WHO wording: “We suggest not to use metformin in the acute management of COVID-19.”	Non-severe COVID-19, regardless of hospitalization risk	Low	Conditional against
Combination antiviral therapy	Do not use combination antiviral therapy outside research or exceptional expert-guided circumstances because evidence is insufficient.	Mild-to-moderate / non-severe COVID-19	Very Low	Conditional against
Systemic corticosteroids	Use systemic corticosteroids for patients with severe or critical COVID-19 requiring oxygen or ventilatory support.	Severe or critical COVID-19	High	Strong

Clinical area	Recommendation	Target population	GRADE evidence quality	Strength
Systemic corticosteroids in non-severe disease	Do not use systemic corticosteroids for non-severe COVID-19 unless there is another medical indication.	Non-severe COVID-19	Moderate	Strong against
IL-6 receptor blockers / JAK inhibitors	Consider immunomodulatory therapy such as IL-6 receptor blockers and/or baricitinib in selected severe or critical COVID-19 patients, usually with corticosteroids.	Severe or critical COVID-19	Moderate	Conditional
Ivermectin	Do not use ivermectin for COVID-19 except in clinical trials.	Any severity	Low	Conditional against
Empirical antibiotics	Use no empirical antibiotics when clinical suspicion of concurrent bacterial infection is low.	Non-severe COVID-19	Low	Strong against empirical antibiotics
Empirical antibiotics	Suggest no empirical antibiotics when clinical suspicion of concurrent bacterial infection is low.	Severe COVID-19	Low	Conditional against empirical antibiotics
Post COVID-19 condition	Provide person-centered, comprehensive, multidisciplinary rehabilitation for persistent functional limitations where needed.	Post COVID-19 condition / long COVID	Low	Conditional

Source basis: WHO therapeutics recommendations and treatment-selection principles are drawn from the August 2025 living guideline and WHO IRIS text [WHO Therapeutics August 2025](#), [WHO IRIS PDF](#). WHO antibiotic recommendations are from the June 2025 clinical management living guideline [WHO Clinical Management June 2025](#), [WHO Clinical Management IRIS PDF](#). Post-COVID condition definition and rehabilitation context are from WHO clinical management and post-COVID materials [WHO Post COVID-19 condition page](#), [WHO fact sheet](#).

1.3 Guideline Development Methodology

Evidence review process

WHO describes the COVID-19 therapeutics guideline as a **living guideline** that dynamically incorporates new evidence, with updates triggered when sufficient evidence is available for the Guideline Development Group to evaluate a therapeutic option [WHO IRIS PDF](#). The August 2025 therapeutic guideline was supported by **living network meta-analyses** and a recent meta-analysis on heparin [WHO IRIS PDF](#).

WHO's clinical management programme similarly states that COVID-19 clinical guidance is based on an ongoing "living" assessment of global evidence, including supportive care and therapeutics [WHO clinical management page](#).

Consensus method

WHO uses a Guideline Development Group with global experts, frontline clinicians, methodologists, and patient partners. The GDG considers not only clinical efficacy and safety, but also **resource implications, acceptability, feasibility, equity, and human rights** [WHO Therapeutics page](#), [WHO IRIS PDF](#).

The therapeutics GDG convenes regularly to review evidence summaries from international clinical trials, and recommendations undergo peer review and WHO Guideline Review Committee review before publication [WHO Therapeutics page](#).

Conflict of interest management

WHO states that Guideline Development Group members were selected to provide balanced representation by global region, gender, and technical expertise. Declarations of interest were collected and managed; competing interests were assessed, and no relevant conflicts were identified in the June 2025 clinical management guideline process [WHO Clinical Management IRIS PDF](#).

Stakeholder and equity considerations

WHO's youth-engagement evidence brief, although not COVID-specific, highlights the broader WHO principle that guideline development should incorporate equity and stakeholder values. The brief reports that the 2022 WHO abortion care guideline integrated youth input through consultations and youth-led convening, and WHO presents this as a precedent for more inclusive guideline development [WHO publication page](#).

II. Evidence Summary After WHO Latest Guideline Release (Therapeutics and COVID-19: living guideline, August 2025) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search strategy

Evidence was reviewed for the period after the WHO August 2025 therapeutic update, with particular attention to sources available through **29 April 2026**.

Core search concepts:

- COVID-19 diagnosis, treatment, management
- WHO, IDSA, CDC, BMJ, PubMed
- Nirmatrelvir/ritonavir, remdesivir, molnupiravir, metformin
- Corticosteroids, IL-6 inhibitors, JAK inhibitors, heparin
- Vaccination, post-COVID condition
- Systematic review, randomized trial, meta-analysis, GRADE

Databases and sources searched

Source type	Sources identified
WHO official guidance	WHO therapeutics, clinical management, post-COVID condition, youth-engagement evidence brief
Specialty society guidelines	IDSA COVID-19 treatment and management updates
Public health agencies	CDC 2025–2026 vaccination guidance
Evidence summaries	BMJ Best Practice COVID-19 diagnosis and management content
Bibliographic databases	PubMed-indexed IDSA guideline updates
Systematic reviews / meta-analyses	BMJ living systematic review and network meta-analysis cited by BMJ Best Practice; broad network meta-analysis summarized in available evidence sources

Inclusion criteria

- Published or current between WHO August 2025 release and 29 April 2026, or cited as current by post-release evidence summaries
- Human COVID-19 diagnosis, treatment, vaccination, or clinical management
- Guidelines using GRADE or systematic evidence review methods
- RCTs, systematic reviews, network meta-analyses, living evidence syntheses, or authoritative guidance

Exclusion criteria

- Non-human studies
- Non-COVID respiratory viral guidance unless directly relevant
- Non-authoritative opinion pieces
- Sources lacking actionable clinical content
- Preprints not reflected in guideline or evidence-synthesis sources

2.2 Key New Studies and Guidance

Study / guidance	Design	Key findings	GRADE quality	Strength / clinical implication
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Study / guidance	Design	Key findings	GRADE quality	Strength / clinical implication
IDSA focused guideline update, published online 1 April 2026: antiviral treatment for mild-to-moderate COVID-19 in adults	GRADE-based clinical practice guideline update using systematic literature review	Provides 9 updated recommendations for nirmatrelvir/ritonavir, remdesivir, and molnupiravir in adults with mild-to-moderate COVID-19 PubMed	Moderate overall for key outpatient antiviral decisions	Supports maintaining early antiviral treatment for patients at increased or high risk
IDSA online guideline: nirmatrelvir/ritonavir	Living guideline / clinical practice guidance	States nirmatrelvir/ritonavir has high efficacy, is oral, and should be the first option considered for ambulatory mild-to-moderate COVID-19; emphasizes drug–drug interaction assessment IDSA	Moderate	Conditional to strong depending on risk group
IDSA online guideline: remdesivir	Living guideline / clinical practice guidance	Recommends IV remdesivir over no antiviral treatment for high-risk mild-to-moderate adults; suggests it for increased-but-not-high-risk adults IDSA	Moderate for high risk; Low for increased-but-not-high risk	Strong for high-risk adults; Conditional for increased-risk adults
IDSA online guideline: molnupiravir	Living guideline / clinical practice guidance	Molnupiravir should not be selected solely for convenience before assessing whether nirmatrelvir/ritonavir or remdesivir can be used IDSA	Low	Conditional; lower-priority alternative

Study / guidance	Design	Key findings	GRADE quality	Strength / clinical implication
BMJ Best Practice diagnosis and management	Evidence-based clinical summary	Suspected or confirmed cases should be isolated, triaged, severity-assessed, and confirmed with RT-PCR when appropriate; rapid antigen tests may be used in selected circumstances BMJ Best Practice PDF	Moderate for RT-PCR diagnostic confirmation; Low to Moderate for antigen-test use depending on context	Strong for diagnostic confirmation and severity triage
BMJ Best Practice on nirmatrelvir/ritonavir	Evidence-based clinical summary citing living systematic review and network meta-analysis	Nirmatrelvir/ritonavir is probably the best treatment for reducing hospital admission compared with standard care, with moderate-certainty evidence; may not reduce symptom duration BMJ Best Practice	Moderate	Supports prioritizing nirmatrelvir/ritonavir in eligible high-risk outpatients
BMJ Best Practice on corticosteroids	Evidence-based clinical summary	Systemic corticosteroids are not recommended in non-severe COVID-19 unless another indication exists, because they may increase mortality risk BMJ Best Practice	Moderate	Strong against in non-severe disease

Study / guidance	Design	Key findings	GRADE quality	Strength / clinical implication
IDSA severe/critical COVID-19 updates, 2026	GRADE-based guideline updates indexed in PubMed	Updates address pemivibart for pre-exposure prophylaxis, vilobelimab for critical illness, and abatacept or infliximab for severe or critical illness PubMed	Low to Moderate, intervention-specific	Does not overturn WHO core recommendations; supports monitoring for immunomodulator-specific updates
CDC 2025–2026 vaccination guidance	Public health recommendation	COVID-19 vaccination is recommended through individual-based or shared clinical decision-making; benefit is most favorable for people at increased risk for severe COVID-19 CDC	Moderate for prevention of severe outcomes; varies by age/risk group	Conditional / shared decision-making
CDC vaccine product and schedule guidance	Public health guidance	Recommended vaccine and number of doses depend on age and vaccination history; available products include Moderna, Pfizer-BioNTech, and Novavax formulations for specified age groups CDC	Moderate	Strong implementation guidance where vaccination is chosen

Study / guidance	Design	Key findings	GRADE quality	Strength / clinical implication
Broad network meta-analysis summarized in available evidence	Systematic review and network meta-analysis; 187 RCTs and 166,230 participants reported	No drug convincingly differed from standard care for mortality, mechanical ventilation, venous thromboembolism, or clinically important bleeding; nirmatrelvir/ritonavir and molnupiravir showed little or no mortality benefit with low certainty Medscape summary	Low to Moderate depending on outcome	Reinforces focusing treatment on patients most likely to benefit and on hospitalization prevention rather than mortality benefit in low-risk populations

2.3 Evidence Gaps and Future Research Needs

Evidence gap	Clinical relevance	Priority	GRADE certainty of current evidence
Direct head-to-head comparisons of nirmatrelvir/ritonavir vs remdesivir vs molnupiravir	Current comparisons are largely indirect; WHO notes low-certainty indirect evidence for comparative effects WHO IRIS PDF	High	Low
Antiviral effectiveness in highly immune populations and newer variants	Baseline hospitalization and mortality risks have changed substantially	High	Low to Moderate
Best treatment thresholds for “increased but not high risk” outpatients	Absolute benefit may be small in lower-risk vaccinated populations	High	Moderate
Role of antiviral therapy in hospitalized but non-critical patients	Evidence varies by timing, severity, and oxygen requirement	Moderate	Low to Moderate
Optimal sequencing or combination of immunomodulators in severe/critical COVID-19	WHO notes trial evidence for adding baricitinib to IL-6 receptor blockers and corticosteroids, but patient selection remains uncertain WHO IRIS PDF	High	Moderate

Evidence gap	Clinical relevance	Priority	GRADE certainty of current evidence
Post-COVID condition treatment	WHO is developing clinical practice guidelines; current management is largely supportive, rehabilitative, and multidisciplinary WHO Post COVID-19 condition page	High	Low
Antibiotic stewardship in severe COVID-19	WHO recommends or suggests no empirical antibiotics when bacterial co-infection suspicion is low, but implementation varies WHO Clinical Management June 2025	High	Low
Pediatric COVID-19 therapeutic thresholds	Children often have milder disease, but severe disease and inflammatory syndromes occur; high-quality trial evidence is limited	Moderate	Low
Vaccine policy under shared clinical decision-making	Risk–benefit balance differs by age, comorbidity, immune status, and prior immunity CDC	Moderate	Moderate

2.4 Updated Recommendations Based on New Evidence

Original WHO-based recommendation	New evidence impact through 29 April 2026	Updated recommendation
Use nirmatrelvir/ritonavir for high-risk non-severe COVID-19 when feasible.	IDSA 2026 update supports continued prioritization; BMJ Best Practice cites moderate-certainty evidence that it probably best reduces hospital admission PubMed , BMJ Best Practice .	Maintain. Use early in eligible high-risk outpatients after drug–drug interaction and renal-function review.
Consider nirmatrelvir/ritonavir for moderate-risk non-severe COVID-19.	IDSA suggests nirmatrelvir/ritonavir over no antiviral treatment for increased-but-not-high-risk adults, conditional with moderate certainty IDSA .	Maintain with individualized decision-making. Prioritize when absolute risk of hospitalization is clinically meaningful.
Consider remdesivir for high-risk non-severe COVID-19 when oral antivirals are unsuitable.	IDSA recommends IV remdesivir for high-risk mild-to-moderate adults, strong with moderate certainty IDSA .	Strengthen operationally. Use as preferred alternative when nirmatrelvir/ritonavir is contraindicated or interactions are unmanageable, if IV access/logistics allow.

Original WHO-based recommendation	New evidence impact through 29 April 2026	Updated recommendation
Use molnupiravir only when preferred antivirals cannot be used.	IDSA advises molnupiravir should not be used solely for convenience before assessing nirmatrelvir/ritonavir or remdesivir feasibility IDSA .	Maintain as lower-priority alternative. Reserve for patients without better antiviral options.
Do not use metformin for acute COVID-19 management.	No post-WHO evidence identified that overturns the WHO conditional recommendation against metformin WHO IRIS PDF .	Maintain. Do not initiate metformin specifically for acute COVID-19 treatment. Continue chronic metformin only for usual indications unless clinically contraindicated.
Avoid systemic corticosteroids in non-severe COVID-19 unless otherwise indicated.	BMJ Best Practice confirms guidelines do not recommend corticosteroids in non-severe disease because they may increase mortality risk BMJ Best Practice .	Maintain strong recommendation against.
Use systemic corticosteroids for severe/critical COVID-19.	No new evidence identified that overturns corticosteroids as backbone therapy in severe disease.	Maintain. Use in patients requiring oxygen or ventilatory support unless contraindicated.
Consider IL-6 receptor blockers and/or JAK inhibitors in selected severe/critical patients.	IDSA notes uncertainty about which critically ill corticosteroid-treated subgroups benefit from additional IL-6 or JAK inhibitors IDSA .	Maintain but refine. Use in selected patients with severe systemic inflammation or progression despite standard care; avoid indiscriminate use.
No empirical antibiotics in non-severe COVID-19 when suspicion of bacterial infection is low.	Stewardship evidence remains supportive; bacterial co-infection rates are generally lower than empirical antibiotic use rates WHO Clinical Management June 2025 , PMC article .	Maintain strong recommendation.
Suggest no empirical antibiotics in severe COVID-19 when suspicion of bacterial infection is low.	No new evidence identified to overturn WHO June 2025 antibiotic stewardship guidance.	Maintain conditional recommendation. Reassess if bacterial infection signs, cultures, imaging, or biomarkers support co-infection.

Original WHO-based recommendation	New evidence impact through 29 April 2026	Updated recommendation
Diagnose suspected COVID-19 with clinical assessment plus confirmatory testing when indicated.	BMJ Best Practice supports immediate isolation, standardized triage, severity assessment, and RT-PCR testing; rapid antigen tests may be used in selected settings BMJ Best Practice PDF .	Add operational diagnostic recommendation. Test early to enable time-sensitive antivirals and infection-control decisions.
Vaccination remains a prevention strategy.	CDC 2025–2026 guidance uses individual-based/shared decision-making, with greatest favorable risk–benefit in people at increased risk for severe COVID-19 CDC .	Add prevention recommendation. Discuss 2025–2026 vaccination using shared decision-making, especially for older adults, immunocompromised people, and those with comorbidities.

Practical WHO-Based Clinical Algorithm as of 29 April 2026

A. Diagnosis and initial assessment

1. **Isolate and triage** suspected or confirmed cases.
2. Assess:
 - Symptom onset date
 - Oxygen saturation
 - Respiratory distress
 - Age, pregnancy, immune status, comorbidities
 - Renal and hepatic function
 - Current medication list for drug–drug interactions
3. Confirm infection with **RT-PCR** when indicated; use rapid antigen testing where appropriate for access, speed, and public health needs [BMJ Best Practice PDF](#).

B. Non-severe COVID-19

Patient group	Preferred approach
Low risk of hospitalization	Symptomatic care; no routine antiviral; avoid corticosteroids and antibiotics unless another indication exists.
Increased or moderate risk	Consider early nirmatrelvir/ritonavir after interaction review; consider remdesivir if oral therapy unsuitable.
High or highest risk	Treat early. Prefer nirmatrelvir/ritonavir if safe; use remdesivir when nirmatrelvir/ritonavir is contraindicated or unsuitable; reserve molnupiravir for when preferred options cannot be used.

C. Severe or critical COVID-19

1. Provide oxygen and organ-supportive care according to severity.

2. Use systemic corticosteroids unless contraindicated.
3. Consider IL-6 receptor blocker and/or JAK inhibitor in selected patients with severe inflammatory disease or progression.
4. Avoid routine empirical antibiotics when suspicion of bacterial infection is low; obtain cultures and reassess if co-infection is suspected.
5. Do not use ivermectin or metformin for acute COVID-19 treatment outside appropriate research contexts.

D. Post COVID-19 condition

Use WHO's clinical definition: symptoms usually begin or persist **3 months from COVID-19 onset**, last at least **2 months**, and are not explained by another diagnosis. Common symptoms include fatigue, shortness of breath, and cognitive dysfunction; symptoms may fluctuate or relapse [WHO Clinical Management IRIS PDF](#). Provide person-centered, multidisciplinary rehabilitation when functional limitation persists [WHO Post COVID-19 condition page](#).

Bottom Line

As of **29 April 2026**, the latest identified WHO COVID-19 therapeutic guideline remains **WHO's Therapeutics and COVID-19: living guideline, August 2025**. The most important WHO update is a **conditional recommendation against metformin** for acute management of non-severe COVID-19. WHO's June 2025 clinical management guideline also added antibiotic-stewardship recommendations against empirical antibiotics when suspicion of bacterial co-infection is low [WHO Therapeutics August 2025](#), [WHO Clinical Management June 2025](#).

Post-WHO-release evidence through **29 April 2026**, especially the **IDSA 2026 antiviral guideline update**, supports continuing a risk-stratified outpatient antiviral strategy: **nirmatrelvir/ritonavir first when feasible, remdesivir as the main alternative**, and **molnupiravir only when preferred options cannot be used** [PubMed](#), [IDSA](#).